

Problem Solving

1. Let $Y_1 < Y_2 < \cdots < Y_n$ be the order statistics of a random sample of size n from $\text{Uniform}(0, 1)$. Show that the k th order statistic Y_k has a beta distribution with parameter $\alpha = k$ and $\beta = n - k + 1$.

2. Let $Y_1 < Y_2$ denote the order statistics of a random sample of size 2 from $N(0, \sigma^2)$. Show that $E(Y_1) = -\sigma/\sqrt{\pi}$.

3. Let $Y_1 < Y_2 < Y_3$ be the order statistics of a random sample of size 3 from a distribution having the pdf $f(x) = 2x$, $0 < x < 1$, zero elsewhere. Show that $Z_1 = Y_1/Y_2$, $Z_2 = Y_2/Y_3$, and $Z_3 = Y_3$ are mutually independent.

